Minds That Meet

A short primer on the philosophical and psychological ideas behind the mutual recognition model, with a reading list

What this is. The mutual recognition model is a small computer simulation. It asks whether some kinds of awareness can only grow *between* two minds, through each one recognizing the other, and whether that growth comes smoothly or in steps. The question is old. This primer sketches where it comes from, in four strands: German Idealism, phenomenology, social theories of the self, and research on infants. It ends with a reading list.

A caution. The model is a toy. It borrows a handful of ideas from these traditions and makes them precise enough to test. It does not interpret any of these thinkers, and the links drawn below are analogies, not claims about what the authors meant. Passages are paraphrased except for a few short, well-known lines.

1. The question

We usually picture self-awareness as something each person has privately, and only afterwards brings into contact with others. A long line of thinkers has argued the reverse. On this view, a mind becomes aware of itself as a mind only through being treated as one by another mind, and treating it in return. Awareness of self and awareness of others grow together, and they grow out of exchange.

If that's right, several further questions follow. What exactly has to happen between two minds for recognition to occur? Does it have to be mutual? Can it be faked? Does it build on itself? These are the questions the model was built to make concrete.

2. German Idealism: the summons and mutual recognition

Fichte: being summoned

Johann Gottlieb Fichte (1762–1814) asked how a finite being could ever come to see itself as free, that is, as a source of its own actions. His answer, in the *Foundations of Natural Right* (1796–97), is that it can't do so alone. It must first be *summoned* (*aufgefordert*) by another free being: addressed as someone who can respond, and left room to do so. The summons is an invitation, not a push. It only works if the one summoned can take it up or not. In answering, the summoned being discovers itself as free, and recognizes the summoner as free too.

> **IN THE MODEL** Minds take turns making *offers*, which the partner may answer in kind or decline. Recognition needs both minds to initiate. A partner who only ever responds can't satisfy it.

Hegel: recognition must be mutual

G. W. F. Hegel (1770–1831) made recognition (*Anerkennung*) central in the *Phenomenology of Spirit* (1807). In the chapter on self-consciousness he argues that a self-conscious being finds the confirmation it seeks only in another self-conscious being: "self-consciousness achieves its satisfaction only in another self-consciousness" (¶175, Miller translation). Things can't supply it, because only another subject can recognize you *as* a subject. Full recognition is therefore reciprocal: at its goal, the two "recognize themselves as mutually recognizing one another" (¶184).

The famous passage on lordship and bondage (the "master–slave dialectic", ¶¶178–196) dramatizes what happens when recognition runs one way. The master is acknowledged by the servant but doesn't acknowledge the servant in return. The result is unsatisfying for both: the master is recognized only by someone he doesn't count as an equal. Hegel's broader point is that one-sided recognition is defective recognition.

> **IN THE MODEL** Recognition requires *mutuality*: both minds must pass the test in the same stretch of time. A "mirror" control that answers perfectly but never initiates is, loosely, the servant's position, and it is never recognized.

3. Phenomenology: how do we experience other minds at all?

Phenomenology, founded by Edmund Husserl (1859–1938), studies experience as it is lived, from the first-person point of view. That raises a hard problem. I never experience your experiences directly. So how do you show up for me as a *subject* rather than as a moving body?

In the Fifth of his *Cartesian Meditations* (1931), Husserl answers with two linked ideas. First, another's inner life is *appresented*: given along with what I perceive, without being perceived itself, much as the hidden back of an object is given along with its visible front. Second, *pairing*: your body and mine are grasped together as alike, and through that likeness I take your body as the expression of a life like mine. The understanding of others is thus not an inference I make after the fact; it is built into how bodies show up to each other.

Later phenomenologists pushed this toward the body and toward infancy. Maurice Merleau-Ponty (1908–1961), in his lectures on the child's relations with others, describes an early stage in which the infant's own body and the bodies of others form a single system before self and other are sorted out.

> **IN THE MODEL** Two minds are analyzed as a single joint system, the *product* of their two graphs. What they can do together depends on how their structures pair up, not on either alone.

4. The social self: Mead and Vygotsky

Two thinkers outside the phenomenological tradition gave the same insight a developmental and empirical shape.

George Herbert Mead (1863–1931), an American pragmatist, argued in *Mind, Self, and Society* (1934, compiled from lecture notes) that the self arises in social interaction. It begins with a *conversation of gestures*: one creature's act calls out a response in another, which calls out a

further response. A gesture becomes meaningful when the one making it can anticipate the response it calls out, which Mead describes as *taking the attitude of the other*. A self, for Mead, is what you get when you can take that attitude toward yourself.

Lev Vygotsky (1896–1934), a Soviet psychologist, put it as a law of development. In *Mind in Society* (1978) he writes that every function in a child's cultural development appears twice: first between people, and later inside the child. Abilities begin as joint activity and are then internalized. His related idea of the *zone of proximal development* describes what a learner can do with help from someone more capable but not yet alone.

> **IN THE MODEL** What a mind "knows" at each level is a record of an exchange it took part in. Vygotsky's more-capable partner corresponds to the model's *elder* experiment, in which newcomers learn from minds further ahead.

5. What infants show

From the 1970s, developmental psychologists began filming parents and young infants face to face, and found that something like a conversation is going on long before language.

Proto-conversation. Mary Catherine Bateson coined the term for the turn-taking exchanges of coos, smiles and vocal sounds between mothers and infants. Colwyn Trevarthen described such exchanges at around two months as *primary intersubjectivity*: the infant isn't just reacting but taking part, with timing and initiative of its own. Public-health writing now calls this *serve and return*.

The still face. In the still-face procedure of Edward Tronick and colleagues (1978), a parent who has been interacting normally suddenly holds a neutral, unresponsive face. Infants of a few months typically try to re-engage, then withdraw and become distressed. What upsets them isn't the parent's absence but the parent's unresponsiveness.

The replay. Lynne Murray and Colwyn Trevarthen (1985) linked two-month-olds and their mothers by live video. Then they replayed a recording of the same mother from moments earlier. The mother's behavior was just as warm, but no longer timed to the infant's, and the infants became withdrawn or distressed. The finding is that infants respond to *contingency*, whether a partner's behavior depends on their own, and not just to warmth. Later studies refined the design (Nadel et al., 1999), and the size and robustness of the effect have been debated, but contingency detection in early infancy is well established (Watson, 1972, is a classic early study).

> **IN THE MODEL** Recognition requires *contingency*: the partner's answers must depend on what was offered. The "recording" control replays a real mind's moves without regard to the partner, a direct analogue of the replay condition, and it was recognized only very rarely, by chance.

6. Closer to the model: interaction as the unit

Recent work in the "enactive" approach to cognition argues that social understanding lives partly in the interaction itself, not only in each individual's head. Hanne De Jaegher and Ezequiel Di Paolo call this *participatory sense-making* (2007).

The closest experimental cousin of the model is *perceptual crossing* (Auvray, Lenay and Stewart, 2009). Two people, each moving a cursor along a shared virtual line, feel a buzz whenever they touch something. Some things they can touch are the partner; others are a static object or a moving decoy that copies the partner's movements. The touch is identical in each case, yet people pick out the real partner, because only a real partner can also be touching *you*, and that mutual contact is what stabilizes. Di Paolo, Rohde and Iizuka (2008) reproduced this with simulated agents. These are the most useful precedents for anyone who wants to see how others have modeled recognition formally.

Finally, the model's *levels* echo the layered structure of shared awareness: we noticed each other; I see that you noticed me; I see that you see that I noticed you. Philosophers describe this layering as *common knowledge*, and Michael Tomasello's work on *shared intentionality* argues it is distinctive of human communication.

A note on "private language." The model's private languages are shared routines that work between two partners who built them and fail with strangers, like inside jokes or the invented languages some twins develop. This is *not* what philosophers usually mean by the phrase. Wittgenstein's private language argument (*Philosophical Investigations*, §243 ff.) concerns a language only one person could understand, and argues that no such language is possible. The model's languages are private to a pair, not to a person, and so are compatible with his argument.

7. What the model tests, and what it found

Idea	How the model makes it precise	What the simulations found
Recognition needs initiative from both sides (Fichte, Hegel)	A *joint cycle*: I offer, you accept; you offer, I accept	Mirrors, which answer perfectly but never initiate, were never recognized in over 450,000 checks
Responsiveness, not just activity (infant replay studies)	A *contingency* score: do your answers depend on my offers?	Recordings fooled minds about 6 times in 100,000 checks, by chance
Development in stages	Nested records: each level contains the one below	Recognition does come in discrete steps
A threshold of responsiveness	A dial for how often minds answer in kind	A sharp tipping point at about 31%, but it lives in each encounter, not in the group as a whole
Growth feeds on itself	Do higher levels make the next one easier?	**No.** Each step was harder than the last, even for the same minds
Shared history makes shared understanding	Compare partners who climbed together with equally advanced strangers	**Yes.** Shared routines worked about 3.7 times as often with the partner they were built with

So the model agrees with the tradition on mutuality, contingency and stages, and produces partner-specific shared structure, but it gives no support to the idea that recognition snowballs. Results depend on modeling choices, and the project report discusses which ones matter.

8. Reading list

START HERE: ACCESSIBLE AND SHORT

Gallagher, S., & Zahavi, D. (2021). *The Phenomenological Mind* (3rd ed.). Routledge. Chapter 9, "How we know others", is the clearest short introduction to the problem of other minds in phenomenology and in cognitive science.

Reddy, V. (2008). *How Infants Know Minds*. Harvard University Press. A readable, argued case that infants understand others through engagement, not detached observation.

Honneth, A. (1995). *The Struggle for Recognition* (J. Anderson, Trans.). Polity Press / MIT Press. (German original 1992.) The standard modern reconstruction of Hegel's idea of recognition, connected to Mead and to psychology.

PRIMARY SOURCES: WHAT TO READ IN EACH

Fichte, J. G. (2000). *Foundations of Natural Right* (F. Neuhouser, Ed.; M. Baur, Trans.). Cambridge University Press. (Original 1796–97.) §3, on the summons. Dense; read it with Williams (below).

Hegel, G. W. F. (1977). *Phenomenology of Spirit* (A. V. Miller, Trans.). Oxford University Press. Or T. Pinkard's 2018 translation, Cambridge University Press. (Original 1807.) Chapter IV, section A, ¶¶166–196, on self-consciousness and lordship and bondage. Very difficult. A commentary helps (Pippin, below).

Husserl, E. (1960). *Cartesian Meditations* (D. Cairns, Trans.). Martinus Nijhoff. (Original 1931.) Fifth Meditation, especially §§50–51 on appresentation and pairing.

Merleau-Ponty, M. (1964). The child's relations with others (W. Cobb, Trans.). In J. M. Edie (Ed.), *The Primacy of Perception* (pp. 96–155). Northwestern University Press. Lectures on intersubjectivity in childhood. More approachable than Husserl.

Mead, G. H. (1934). *Mind, Self, and Society* (C. W. Morris, Ed.). University of Chicago Press. Part II, "Mind", on gestures and significant symbols; Part III, "The Self", on taking the attitude of the other.

Vygotsky, L. S. (1978). *Mind in Society* (M. Cole, V. John-Steiner, S. Scribner, & E. Souberman, Eds.). Harvard University Press. Chapter 4, on internalization ("every function appears twice"), and chapter 6, on the zone of proximal development. Short and readable.

INFANT RESEARCH

Bateson, M. C. (1975). Mother–infant exchanges: The epigenesis of conversational interaction. *Annals of the New York Academy of Sciences, 263*, 101–113. https://doi.org/10.1111/j.1749-6632.1975.tb41575.x

Trevarthen, C. (1979). Communication and cooperation in early infancy: A description of primary intersubjectivity. In M. Bullowa (Ed.), *Before Speech* (pp. 321–347). Cambridge University Press.

Tronick, E., Als, H., Adamson, L., Wise, S., & Brazelton, T. B. (1978). The infant's response to entrapment between contradictory messages in face-to-face interaction. *Journal of the American Academy of Child Psychiatry, 17*(1), 1–13. The original still-face study.

Murray, L., & Trevarthen, C. (1985). Emotional regulation of interactions between two-month-olds and their mothers. In T. M. Field & N. A. Fox (Eds.), *Social Perception in Infants* (pp. 177–197). Ablex. The live-versus-replay video study.

Nadel, J., Carchon, I., Kervella, C., Marcelli, D., & Réserbat-Plantey, D. (1999). Expectancies for social contingency in 2-month-olds. *Developmental Science, 2*(2), 164–173. https://doi.org/10.1111/1467-7687.00065 A later, tighter version of the replay design.

Watson, J. S. (1972). Smiling, cooing, and "the game." *Merrill-Palmer Quarterly, 18*(4), 323–339. An early study of infants' sensitivity to contingency.

COMMENTARY AND CONTEMPORARY WORK

Williams, R. R. (1992). *Recognition: Fichte and Hegel on the Other*. SUNY Press.

Pippin, R. B. (2011). *Hegel on Self-Consciousness: Desire and Death in the Phenomenology of Spirit*. Princeton University Press. A short book on exactly the chapter that matters here.

Zahavi, D. (2014). *Self and Other*. Oxford University Press.

De Jaegher, H., & Di Paolo, E. (2007). Participatory sense-making: An enactive approach to social cognition. *Phenomenology and the Cognitive Sciences, 6*(4), 485–507. https://doi.org/10.1007/s11097-007-9076-9

Auvray, M., Lenay, C., & Stewart, J. (2009). Perceptual interactions in a minimalist virtual environment. *New Ideas in Psychology, 27*(1), 32–47. https://doi.org/10.1016/j.newideapsych.2007.12.002 The perceptual crossing experiment.

Di Paolo, E. A., Rohde, M., & Iizuka, H. (2008). Sensitivity to social contingency or stability of interaction? Modelling the dynamics of perceptual crossing. *New Ideas in Psychology, 26*(2), 278–294. https://doi.org/10.1016/j.newideapsych.2007.07.006 The nearest precedent for simulating recognition with artificial agents.

Tomasello, M. (2008). *Origins of Human Communication*. MIT Press. On shared intentionality and layered mutual awareness.

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